# A Priori Error Analysis

Galerkin orthogonality, best approximation, and Céa's lemma

# Lecture outline

The finite element computation gives $u_h$. We now ask how $u_h$ compares with the exact weak solution $u$.

1. State the continuous and discrete variational problems and their assumptions.
2. Introduce the error norms and relate the energy and $H^1$ norms.
3. Derive Galerkin orthogonality and energy-norm best approximation.
4. State and prove Céa's lemma.

The next lecture introduces finite element interpolation and uses it to obtain error estimates in terms of the mesh size $h$.

# Continuous and discrete problems

Let

$$\mathcal{V}_0 = H^1_{0,\Gamma_D}(\Omega), \qquad a(w,v) = \int_\Omega k\,\nabla w \cdot \nabla v \,\mathrm{d}x.$$

The continuous problem is

$$\boxed{\text{find } u \in \mathcal{V}_0 : \quad a(u,v) = \ell(v) \quad \forall v \in \mathcal{V}_0.}$$

For a conforming finite element space $\mathcal{V}_{h,0} \subset \mathcal{V}_0$, the discrete problem is

$$\boxed{\text{find } u_h \in \mathcal{V}_{h,0} : \quad a(u_h,v_h) = \ell(v_h) \quad \forall v_h \in \mathcal{V}_{h,0}.}$$

The two problems use the same bilinear and linear forms. Only the spaces differ.

# Assumptions for the analysis

We assume throughout this lecture that

- $\Omega$ is a bounded Lipschitz domain and $\Gamma_D$ has positive boundary measure;
- $0 < k_0 \leq k(\boldsymbol{x}) \leq k_1$ almost everywhere in $\Omega$;
- $\mathcal{T}_h$ is conforming and resolves the boundary partition;
- $\mathcal{V}_{h,0}$ is a continuous Lagrange space of degree $p \geq 1$ with homogeneous values on $\Gamma_D$; and
- the forms $a$ and $\ell$ are evaluated exactly.

If quadrature or boundary approximation changes the discrete forms, the orthogonality relation contains an additional consistency error. We do not consider that case in this lecture.

# Continuity and coercivity

The analysis uses two estimates:

$$|a(w,v)| \leq M \|w\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}, \qquad w, v \in \mathcal{V}_0,$$

and

$$a(v,v) \geq \alpha \|v\|^2_{H^1(\Omega)}, \qquad v \in \mathcal{V}_0.$$

For scalar diffusion,

$$M = k_1, \qquad \alpha = \frac{k_0}{1 + C_P^2}, \qquad \frac{M}{\alpha} = \frac{k_1}{k_0}(1 + C_P^2).$$

The constants depend on the coefficient and the domain, but not on the mesh size $h$ or polynomial degree $p$.

# Existence and stability of the discrete solution

Because $\mathcal{V}_{h,0}$ is a finite-dimensional subspace of $\mathcal{V}_0$, the same continuity and coercivity estimates hold on $\mathcal{V}_{h,0}$.
The Lax–Milgram theorem therefore gives a unique finite element solution and

$$\|u_h\|_{H^1(\Omega)} \leq \frac{1}{\alpha} \|\ell\|_{\mathcal{V}_0^*} .$$

Coercivity also gives a positive-definite stiffness matrix. For any nonzero coefficient vector $\boldsymbol{V}$ representing $v_h \in \mathcal{V}_{h,0}$,

$$\boldsymbol{V}^T \boldsymbol{K}_{ff} \boldsymbol{V} = a(v_h, v_h) \geq \alpha \|v_h\|^2_{H^1(\Omega)} > 0.$$

When $a$ is symmetric, the reduced stiffness matrix $\boldsymbol{K}_{ff}$ is symmetric positive definite.

# Measures of the error

Define

$$e = u - u_h.$$

Because $u, u_h \in \mathcal{V}_0$, the error also belongs to $\mathcal{V}_0$.

$$\|e\|_{L^2(\Omega)} \quad \text{measures error in the solution values,}$$

$$|e|_{H^1(\Omega)} = \|\nabla e\|_{L^2(\Omega;\mathbb{R}^d)} \quad \text{measures error in the gradient,}$$

$$\|e\|_{H^1(\Omega)} = \left(\|e\|^2_{L^2(\Omega)} + |e|^2_{H^1(\Omega)}\right)^{1/2} \quad \text{measures both.}$$

For diffusion, the gradient error determines the flux error. For elasticity, it determines the strain error.

# Energy norm

Suppose $a$ is symmetric, continuous, and coercive on $\mathcal{V}_0$. Define

$$\boxed{\|v\|_a = a(v,v)^{1/2}.}$$

This is a norm because $a$ is an inner product on $\mathcal{V}_0$.
For scalar diffusion,

$$\|v\|_a^2 = \int_\Omega k\,|\nabla v|^2 \,\mathrm{d}x.$$

The coefficient bounds give

$$\sqrt{k_0}\;|v|_{H^1(\Omega)} \le \|v\|_a \le \sqrt{k_1}\;|v|_{H^1(\Omega)}\,.$$

The energy norm weights the gradient by the material coefficient $k$.

# Energy norm and the $H^1$ norm

On $\mathcal{V}_0$, the Poincaré inequality gives

$$\|v\|^2_{H^1(\Omega)} \leq (1 + C_P^2)\, |v|^2_{H^1(\Omega)}\,.$$

Combining this with the coefficient bounds gives

$$\boxed{\sqrt{\frac{k_0}{1 + C_P^2}}\ \|v\|_{H^1(\Omega)} \leq \|v\|_a \leq \sqrt{k_1}\ \|v\|_{H^1(\Omega)}\,.}$$

Thus convergence in the energy norm is equivalent to convergence in the $H^1$ norm on $\mathcal{V}_0$. The numerical values of the two norms need not be equal.

# Energy norm for an axial bar

For an axial bar with $v(0) = 0$,

$$a(w, v) = \int_0^L AE\, w' v' \, \mathrm{d}x, \qquad \|e\|_a^2 = \int_0^L AE\, (e')^2 \, \mathrm{d}x.$$

The energy norm weights the strain error by the axial rigidity. When $AE = 1$,

$$\|e\|_a = |e|_{H^1(0,L)}.$$

Because $e(0) = 0$, the Poincaré inequality makes this seminorm equivalent to the full $H^1(0, L)$ norm.

Without an essential boundary condition, a nonzero constant displacement has zero strain and zero value of $a(v, v)$. The energy expression is then only a seminorm until rigid translations are removed.

# Galerkin orthogonality

Every $v_h \in \mathcal{V}_{h,0}$ is also an admissible test function in the continuous problem. Hence

$$a(u, v_h) = \ell(v_h), \qquad a(u_h, v_h) = \ell(v_h).$$

Subtracting the discrete equation from the continuous equation gives

$$\boxed{a(u - u_h, v_h) = 0 \qquad \forall v_h \in \mathcal{V}_{h,0}.}$$

This conclusion uses conformity, $\mathcal{V}_{h,0} \subset \mathcal{V}_0$, and the same forms $a$ and $\ell$ in the continuous and discrete problems. Symmetry of $a$ is not needed for this result.

# Orthogonality for a symmetric problem

If $a$ is also symmetric, then

$$(w, v)_a := a(w, v)$$

is an inner product on $\mathcal{V}_0$, with induced norm

$$\|v\|_a = a(v, v)^{1/2}.$$

With $e = u - u_h$, Galerkin orthogonality becomes

$$(e, v_h)_a = 0 \qquad \forall v_h \in \mathcal{V}_{h,0}.$$

Thus the error is orthogonal to every function in the finite element space, where orthogonality is measured by the inner product $a$.

# Pythagorean identity in the energy norm

For any $v_h \in \mathcal{V}_{h,0}$,

$$u - v_h = (u - u_h) + (u_h - v_h).$$

The second term belongs to $\mathcal{V}_{h,0}$, so Galerkin orthogonality gives

$$a(u - u_h, u_h - v_h) = 0.$$

Using bilinearity and symmetry of $a$,

$$\begin{aligned} \|u - v_h\|_a^2 &= a\big((u - u_h) + (u_h - v_h), (u - u_h) + (u_h - v_h)\big) \\ &= \|u - u_h\|_a^2 + 2a(u - u_h, u_h - v_h) + \|u_h - v_h\|_a^2 . \end{aligned}$$

The middle term is zero by Galerkin orthogonality. Therefore,

$$\boxed{\|u - v_h\|_a^2 = \|u - u_h\|_a^2 + \|u_h - v_h\|_a^2 .}$$

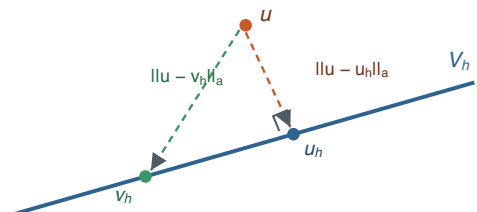
# Best approximation in a selected norm

Let $V_h$ be a finite-dimensional subspace of a normed vector space $V$. The best-approximation error is

$$E_V(u; V_h) = \min_{v_h \in V_h} \|u - v_h\|_V .$$

It is the shortest distance, measured with $\|\cdot\|_V$, from $u$ to a function in $V_h$.

Changing the norm changes how distance is measured and can change which function in $V_h$ is closest to $u$.



# The Galerkin solution is the energy-norm best approximation

For any $v_h \in \mathcal{V}_{h,0}$, the Pythagorean identity gives

$$\|u - v_h\|_a^2 = \|u - u_h\|_a^2 + \|u_h - v_h\|_a^2 .$$

Since the last term is nonnegative,

$$\|u - v_h\|_a \geq \|u - u_h\|_a \qquad \forall v_h \in \mathcal{V}_{h,0}.$$

Equality holds when $v_h = u_h$. Therefore,

$$\boxed{\|u - u_h\|_a = \min_{v_h \in \mathcal{V}_{h,0}} \|u - v_h\|_a .}$$

The finite element solution is the function in $\mathcal{V}_{h,0}$ closest to $u$ in the energy norm.

# A weighted best-approximation example

Let

$$V = \mathbb{R}^2, \qquad V_h = \operatorname{span}\{(1,1)\}, \qquad \boldsymbol{u} = (1,0),$$

and define

$$a(\boldsymbol{x}, \boldsymbol{v}) = 4x_1 v_1 + x_2 v_2, \qquad \|\boldsymbol{x}\|_a = \sqrt{4x_1^2 + x_2^2}.$$

Every vector in $V_h$ has the form $\boldsymbol{v}_h = t(1,1)$. Therefore

$$\|\boldsymbol{u} - \boldsymbol{v}_h\|_a^2 = 4(1-t)^2 + t^2.$$

Differentiating this quadratic with respect to $t$ gives

$$-8(1-t) + 2t = 0, \qquad t = \frac{4}{5}, \qquad \boldsymbol{u}_h = \left(\frac{4}{5}, \frac{4}{5}\right).$$

The Euclidean closest point is $\frac{1}{2}(1,1)$. The discrete space is the same, but the norm changes the best approximation.

# Céa's lemma

Let $V$ be a Hilbert space and $V_h \subset V$ a finite-dimensional subspace. Assume that

$$|a(w,v)| \leq M \|w\|_V \|v\|_V , \qquad a(v,v) \geq \alpha \|v\|_V^2 .$$

If $u$ and $u_h$ solve the continuous and discrete Galerkin problems, then

$$\boxed{\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V .}$$

The estimate compares the computed error with the smallest error attainable in $V_h$.

# Proof of Céa's lemma: coercivity and orthogonality

Let $e = u - u_h$ and choose any $v_h \in V_h$. Coercivity gives

$$\alpha \|e\|_V^2 \leq a(e, e).$$

Because

$$e = (u - v_h) + (v_h - u_h)$$

and $a$ is bilinear in its second argument,

$$a(e, e) = a(e, u - v_h) + a(e, v_h - u_h).$$

Because $v_h - u_h \in V_h$, Galerkin orthogonality gives

$$a(e, v_h - u_h) = 0.$$

Hence

$$\alpha \|e\|_V^2 \leq a(e, u - v_h).$$

# Proof of Céa's lemma: continuity

Continuity of $a$ gives

$$\alpha \|e\|_V^2 \leq M \|e\|_V \|u - v_h\|_V .$$

If $e \neq 0$, divide by $\alpha \|e\|_V$:

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \|u - v_h\|_V .$$

This holds for every $v_h \in V_h$. Taking the infimum gives Céa's estimate.

For diffusion in the $H^1$ norm,

$$\frac{M}{\alpha} = \frac{k_1}{k_0}(1 + C_P^2).$$

This factor does not depend on $h$ or $p$.